

SHIV DUTT CHOUBEY

M.Sc. Cybersecurity Candidate | Cybersecurity Researcher

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Professional Summary

Motivated Cyber Security graduate student and researcher pursuing an M.Sc. in Cyber Security at the National Forensic Sciences University (NFSU), Gandhinagar (CGPA: 9.05). Former NPTEL Pre-Doctoral Fellow at IIT Kharagpur, conducting research on Quantum Internet Architecture and Post-Quantum Networking. Skilled in network security, ethical hacking, cryptography, post-quantum cryptography, IoT security, secure API development, intrusion detection, and AI-driven security solutions. Proficient in programming, cybersecurity risk assessment, and implementing security controls aligned with NIST and ISO/IEC 27001. Passionate about building secure systems, analyzing cyber threats, and solving complex cybersecurity challenges through research and practical innovation.

Core Competencies

- Network Security & IDPS
- Ethical Hacking & Penetration Testing
- Cryptography & Post-Quantum Security
- Intrusion Detection Systems
- MITRE ATT&CK Mapping
- Secure Coding Practices
- Malware Analysis
- Web, Mobile Application Security
- Quantum Computing Research
- Industrial IoT (IIoT) Security
- Natural Language Processing
- API Development & Integration
- Patent Analysis & IP Research
- Database Management
- Cloud Computing Fundamentals
- 5G - RAN
- Secure Federated Learning

Education

Master of Science in Cyber Security <i>National Forensic Sciences University (NFSU), Gandhinagar, Gujarat</i>	Pursuing — July 2027 SPI: 9.05 (Semester 2)
Bachelor of Science in Computer Science — 80 % <i>Rani Durgawati Vishwavidyalaya (RDVV), Jabalpur, Madhya Pradesh</i>	July 2024
AISSCE (Class XII) — CBSE 91 % <i>Army Public School No. 1, Jabalpur, Madhya Pradesh</i>	July 2021
AISSE (Class X) — CBSE 91.2 % <i>Army Public School No. 1, Jabalpur, Madhya Pradesh</i>	July 2019

Professional Experience

NPTEL Pre-Doctoral Research Fellow — *Indian Institute of Technology Kharagpur (IIT-KGP), Kharagpur, West Bengal, India(On-Site), August 2024 – July 2025 (12 months)*

- Conducted applied research in Quantum Computing, Healthcare IoT (IIoT), and AI/ML under the sponsored project titled “Design and Development of Future Quantum Internet Architecture and Post-Quantum Modern Networking.”
- Contributed to the design of post-quantum cryptographic protocols to secure next-generation Internet infrastructure.
- Investigated communication systems specific to IIoT environments and proposed improved frameworks.
- Collaborated with faculty researchers and peer fellows; produced technical reports and contributed to academic deliverables.

NPTEL Summer Research Intern — *RGSOIPL, IIT Kharagpur & K.L.E. Technological University(Remote)*
April 2024 – July 2024 (4 Months)

- Worked at the intersection of Intellectual Property Law and technology, analysing patent landscapes in emerging domains including AI and software development ecosystem security.
- Developed proficiency in patentability searches, prior-art analysis, and technical writing for IP documentation.

Software Developer Intern — *Doions Private Limited (On-Site), November 2022 – March 2023, Jabalpur, Madhya Pradesh, India (5 months)*

- Designed and implemented a **Data Processing Pipeline for Conversational AI** integrated into a production chatbot framework (*Local Learn Chatbot*), improving data throughput and model response quality.
- Participated in regular Agile sprint planning, code reviews, and retrospective meetings; improved team delivery velocity.
- Contributed to backend API development and quality assurance processes for client-facing software products.

Research Projects & Technical Portfolio

ERIV: Post-Decryption Semantic Protection Layer Against Harvest-Now-Decrypt-Later (HNDL) Attacks

- Developed a semantic security layer designed to protect data *after* decryption, targeting the emerging HNDL quantum threat model where adversaries collect encrypted data today for future quantum decryption.

Netflow Monitor — Low-Cost Intrusion Detection & Prevention System (IDPS) for SMBs

- Architected a lightweight, easily-integrable IDPS leveraging network flow analysis, designed specifically for resource-constrained small and medium-sized businesses.
- Implemented anomaly detection algorithms to identify malicious traffic patterns with minimal false-positive rates.

LoRa ImageSum — Secure Image Transmission over LoRa Protocol

- Created a novel technique to securely transmit images over the LoRa (Long Range) low-power radio protocol, incorporating cryptographic checksums to ensure data integrity across lossy channels.

NLP-AST Compiler Security Technique

- Developed a method combining Abstract Syntax Tree (AST) analysis with Natural Language Processing to detect source code manipulation and unauthorised access attempts prior to compilation.

Keyword Detection API

- Built and deployed a RESTful API to detect and flag specific keywords in submitted text, applicable to content moderation and threat-intelligence workflows.

LiveMedQR

- Built a hardware device attachable to any medical unit to transmit data from the medical device to patient's device securely over Wi-Fi to preapre the Electronic Healthcare Record (EHR) of the patient to reduce the errors committed by medical staff to improve the accuracy and correctness of medical data.

Certifications

Forensic Aptitude and Caliber Test(FACT) 2026 - Digital Forensics

Fortinet NSE Training Institute

- **Fortinet Certified Associate — NSE 3** — *Fortinet NSE Training Institute*
Network security architecture, Fortinet product portfolio, threat intelligence
- **Fortinet Certified Fundamentals — NSE 1 & 2** — *Fortinet NSE Training Institute*
Cyber threat landscape, information security fundamentals, cloud security basics

NPTEL — Indian Institute of Technology (IIT)

- **Ethical Hacking** — IIT Kharagpur — TCP/IP, Secure Coding, Cryptography, Malware Analysis, Web Security
- **Applied Artificial Intelligence** — IIT Palakkad — PyTorch, TensorFlow, NLP, LLMs, Distributed Deep Learning
- **Computer Networks & Internet Protocol** — IIT Kharagpur — TCP/IP, Routing, Switching, Network Protocols
- **Introduction to Industry 4.0 & IIoT** — IIT Kharagpur — IIoT Architecture, Cyber-Physical Systems, Industrial Automation
- **Introduction to Internet of Things** — IIT Kharagpur — IoT Protocols, Edge Computing, Sensor Networks
- **Introduction to Database Systems** — IIT Madras — Relational Databases, Query Processing, Database Design
- **Database Management System** — IIT Kharagpur — SQL, NoSQL, Data Modelling, Relational Algebra
- **Programming in Java** — IIT Kharagpur — Java, OOP, Agile Application Development
- **Programming in Modern C++** — IIT Kharagpur — C++, Memory Management, OOP, Debugging

- **An Introduction to Programming through C++** — IIT Bombay — C++, Algorithms, Fundamental Programming
- **Introduction to Operating Systems** — IIT Madras — Linux, OS Internals, System Administration
- **Computer Graphics** — IIT Madras — Graphics Pipeline, OpenGL, Shader Programming, 3D Transformations
- **Corporate Finance** — IIT Kharagpur — Capital Budgeting, Firm Valuation, Risk & Return Analysis
- **Intellectual Property Portfolio Management** — IIT Roorkee — IP Management, Portfolio Analysis
- **Roadmap for Patent Creation** — IIT Kharagpur — Patent Requirement, Patent Application writing, Portfolio Analysis

Technical Skills

Languages: Python, Java, C, C++, Kotlin, JavaScript, SQL, Bash/Shell Scripting

Networking: Network Packet Analysis, Firewall Configuration, VPN, LoRa, Networking device hardening

AI / ML: TensorFlow, PyTorch, NLP pipelines, , Federated Learning, Secure Agentic Workflow

Databases: MySQL, PostgreSQL, NoSQL (MongoDB), database design and administration

Cloud & DevOps: Cloud Computing fundamentals, Docker (basics), Kubernetes, GPU Acceleration

Development: RESTful API design, Agile/Scrum, Git, Full Stack Web and Mobile Application

Digital Forensics: Seizure and Analysis of Electronic Evidences

Security Tools & Frameworks: Wireshark, Nmap, Metasploit, Zed Attack Proxy, VeraCrypt, Frida, Objection

Information Security: Security Controls, Security Baselines, Secure Configuration, Information Risk Management

Competitions, Hackathons & Workshops

- **Quantum Science Conclave 2026** — Participated in a hands-on workshop hosted by **C-DAC Pune** and **IIT Delhi**, gaining practical exposure to Quantum Gates, Grover's Algorithm, Quantum Error Correction, and the **Qniverse** quantum simulation platform.
- **5G Innovation Hackathon 2026** — Selected for National Level Presentation by the **Department of Telecommunications**, Government of India.
- **Anti-Forensics Technique Hackathon** — Designed and demonstrated a method to identify VeraCrypt-encrypted hidden volumes; showcased advanced digital forensics skills.
- **File Forensics Bootcamp** — Completed intensive training in reconstructing fragmented distributed files from forensic disk images.
- **Inter-Group Cyber Security Presentation Competition** — Presented research on emerging phishing and social engineering attack vectors. **Secured 1st Position.**
- **Innovate India Coding Championship** — Solved competitive DSA and aptitude problems in *C++* under timed conditions. **Achieved National Rank 7,845.**

Volunteer Work & Leadership

- **WeSeSo (We Secure Society)** — Cybersecurity Volunteer (April 2020 – Present): Raises public awareness of digital safety, cyber threats, and responsible internet use through outreach activities and community programmes.
- **National Service Scheme (NSS), Government of India** — Student Volunteer (November 2021 – July 2024): Participated in rural development camps, health awareness drives, and community welfare initiatives.
- **National Cadet Corps (NCC), India** — India Cadet, Army Wing (July 2016 – July 2018): Developed leadership, discipline, and teamwork through structured military training activities.